Retrieval-Augmented Generation (RAG) Complete Beginner to Advanced Guide

This PDF provides a complete guide to Retrieval-Augmented Generation (RAG), including embeddings, vector databases, chunking, retrieval pipelines, prompt engineering, production deployment, and real-world applications.

1. Introduction to RAG

Retrieval-Augmented Generation (RAG) is a modern AI architecture that combines retrieval systems with large language models. Instead of depending only on training data, the model retrieves relevant information from external documents and uses that context to answer questions. RAG improves factual accuracy, reduces hallucinations, supports real-time knowledge updates, and enables private enterprise knowledge systems. A typical RAG pipeline includes document ingestion, chunking, embeddings, vector indexing, retrieval, reranking, prompt engineering, and response generation. Modern systems often integrate Pinecone, FAISS, ChromaDB, LangChain, OpenAI embeddings, and transformer-based reranking models.

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2. Why RAG Matters

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3. LLMs and Knowledge Limitations

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4. Document Loading

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5. Chunking Strategies

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6. Embeddings

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10. Retriever Pipelines

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11. Prompt Engineering

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12. RAG Architecture

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13. Hybrid Search

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14. Metadata Filtering

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15. Reranking

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16. Query Expansion

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17. Context Windows

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18. Hallucination Reduction

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19. Conversation Memory

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20. Streaming Responses

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21. Evaluation Metrics

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22. Production Deployment

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24. RAG with LangChain

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25. RAG with LlamaIndex

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26. Fine-Tuning vs RAG

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28. Advanced Retrieval

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29. Real-World Applications

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30. Future of RAG

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Conclusion

RAG is one of the most important architectures in modern AI systems. It allows developers to build accurate, scalable, and domain-aware AI applications using external knowledge sources.